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Inferring Hidden States and Decision-Relevant Information from Neighboring Agents' Actions in Multi-Agent Systems

Zhishan Wang[1†]

[1]School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
(E-mail: zwang103@e.ntu.edu.sg)

Abstract: Multi-agent coordination often relies on explicit communication, yet agents also reveal task-relevant information through their physical behavior. This paper studies whether relations among multiple observed behaviors can reveal information that is unavailable from each behavior alone, and whether such information can improve the observer's own decision. We formulate a behavior-mediated coordination framework that connects local observation, joint inference, mission/role selection, and physical execution. In a football-inspired model, we show that two individually uninformative motions can become jointly informative about a latent tactical opportunity, and derive conditions under which this information changes the observer's mission from waiting to active coverage. Closed-loop simulations show that the resulting decision gain can persist under physical dynamics. A ten-agent formation study further shows that motion-based reasoning improves over delayed reaction, while providing no additional benefit over a strong position-only baseline in the tested settings. These results distinguish behavioral identifiability, decision value, and closed-loop coordination benefit.

Keywords: Multi-agent systems; networked control; distributed estimation; behavior-mediated coordination; decentralized decision-making; distributed control; formation control; sports tactical analysis; team collaboration.

1. INTRODUCTION AND RELATED WORK

Coordination is a central problem in multi-agent systems [1, 2], and many decentralized methods rely on explicit information exchange or consensus over local beliefs [3]. However, even without explicit runtime messages, agents continuously reveal information through their actions, trajectories, and responses to one another. This raises a fundamental question: can relations among multiple observed behaviors reveal task-relevant information that is unavailable from any individual behavior alone? More importantly, can such information change the observer's own mission or role and improve closed-loop coordination? Our focus is therefore not merely whether teammate behavior is interpretable, but whether relational structure among observed behaviors creates information that is valuable for the observer's own decision.

Behavior-based teammate inference has been studied extensively in ad hoc teamwork and cooperative joint action. Melo and Sardinha infer teammates' tasks from observed behavior and adapt the observer's policy accordingly [4], while Barrett and Stone study cooperation with unknown teammates in robot soccer [5]. Maisto et al. further show that agents can infer a joint goal from partners' movements through interactive inference [6]. More recent work considers partially observable and recurrent Bayesian task/team inference [7, 8], while inverse planning provides a broader foundation for inferring latent goals and states from observed actions [9]. These studies establish that behavior can support teammate or task inference followed by adaptation. Our focus is narrower and complementary: can task-relevant information arise specifically from the *relation* between individually ambiguous behaviors, can that information change the observer's own mission or role, and does the resulting decision advantage persist after physical execution?

In this work, we study behavior-mediated coordination through the inference-to-action chain. Locally observed teammate motion provides evidence about hidden task-relevant variables, which is then used to adapt the observer's mission, role, and physical reference. A football-inspired model isolates the key mechanism: two individually uninformative motions can become informative through their relation. We connect this relational identifiability to a WAIT-to-COVER transition and closed-loop physical execution.

The contribution is an explicit link from relational behavioral information to the observer's own decision and its closed-loop consequence. We establish when joint teammate behavior reveals a latent opportunity absent from the individual marginals, derive a mission/role switching condition, and evaluate the resulting action under physical execution. Football simulations closely match the analytical prediction and reduce closed-loop cost. A ten-agent formation study further shows that anticipatory reasoning improves over delayed reaction but not over a strong position-only baseline, thereby separating behavioral identifiability, decision value, and closed-loop benefit.

2. PROBLEM FORMULATION AND FRAMEWORK

2.1. Local Observation and Information

Consider a multi-agent system with agent set $\mathcal{V} = \{1, \ldots, N\}$, where N denotes the number of agents. At discrete time k, agent $i \in \mathcal{V}$ has a physical state $x_i(k) \in \mathcal{X}_i$, a control input $u_i(k) \in \mathcal{U}_i$, a mission state $q_i(k) \in \mathcal{Q}_i$, and a functional role $r_i(k) \in \mathcal{R}_i$. The collective physical state is denoted by $x(k) =$

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$\mathrm{col}(x_1(k), \ldots, x_N(k))$. Let $e(k)$ denote the shared environment state, which may contain task-relevant quantities such as object, target, obstacle, ball, or opponent states. The physical evolution of agent i is represented in the general form

$$x_i(k+1) = f_i\big(x(k), u_i(k), e(k), w_i(k)\big), \tag{1}$$

where f_i is the state-transition map and $w_i(k)$ represents process disturbances or unmodeled physical effects.

Agent i obtains two types of local evidence. First, it receives a private observation

$$Y_i(k) = h_i\big(x(k), e(k), v_i(k)\big), \tag{2}$$

where h_i is the local sensing map and $v_i(k)$ denotes the corresponding private measurement noise. Second, when a teammate $j \neq i$ is locally visible, agent i extracts a physical behavior feature

$$B_{ij}(k) = \mathcal{O}_{ij}\big(x_j(k-1), x_j(k), \nu_{ij}(k)\big), \tag{3}$$

where $\mathcal{O}_{ij}$ is the behavior-feature extraction map and $\nu_{ij}(k)$ denotes the corresponding observation noise. Thus, $B_{ij}(k)$ is constructed from the physical behavior of teammate j observed over the completed interval $(k-1, k]$, and can be used by agent i when making its decision at time k.

Local visibility induces a directed, time-varying observation graph $\mathcal{G}_o(k) = (\mathcal{V}, \mathcal{E}_o(k))$. For $j \neq i$, let $M_{ij}(k) \in \{0, 1\}$ denote the visibility indicator, where $M_{ij}(k) = 1$ means that agent i can observe the physical behavior of teammate j during the corresponding observation interval. Accordingly,

$$(j, i) \in \mathcal{E}_o(k) \quad \Longleftrightarrow \quad M_{ij}(k) = 1. \tag{4}$$

The observable-neighbor set of agent i is defined as

$$\mathcal{N}_i^o(k) = \{j \in \mathcal{V} \setminus \{i\} \,:\, M_{ij}(k) = 1\}. \tag{5}$$

Hence, an edge $j \to i$ indicates that agent i observes the physical behavior of agent j. It does not represent an explicit communication link, and the observation graph need not be symmetric or time invariant.

Let K_i denote the fixed local knowledge base of agent i, including its hypothesis set, behavior model, admissible mission and role libraries, and reference and control rules. Before using teammate behavior, the information available to agent i at time k is represented by

$$\mathcal{F}_i^k = \sigma\big(K_i,\, Y_i^{0:k},\, x_i^{0:k},\, u_i^{0:k-1}\big), \tag{6}$$

where the superscript $0{:}k$ denotes the observation history from time 0 to time k.

To collect the locally observed teammate behavior, define

$$D_i(k) = \big(\{M_{ij}(k)\}_{j \neq i}, \{B_{ij}(k)\}_{j \in \mathcal{N}_i^o(k)}\big), \tag{7}$$

where $D_i(k)$ contains both the current visibility pattern and the behavior features available to agent i. The behavior-augmented information is then

$$\mathcal{G}_i^k = \mathcal{F}_i^k \vee \sigma\big(D_i^{0:k}\big), \tag{8}$$

where $\vee$ denotes the smallest sigma-field containing both $\mathcal{F}_i^k$ and the history of observed teammate behavior.

Throughout this paper, "without explicit communication" means that agents do not exchange runtime messages, beliefs, semantic labels, or raw sensor packets for the coordination decisions under study. Coordination is instead mediated by private sensing and observable physical behavior. Shared prior structure, such as a common coordinate convention, task or role libraries, and the local behavior model encoded in K_i, may be available before the tactical episode begins.

2.2. Behavior-Mediated Inference and Action

Let $\eta \in \mathcal{H}_\eta$ denote a task-relevant latent variable that is not directly available to agent i. For each teammate $j \neq i$, define its mission-role state as $Z_j(k) = (q_j(k), r_j(k))$. The latent state considered by agent i is

$$\Theta_i(k) = \big(\eta, Z_{-i}(k), \chi_i(k)\big), \tag{9}$$

where $Z_{-i}(k) = \{Z_j(k) : j \neq i\}$ and $\chi_i(k)$ collects any additional unobserved context required by the local inference model. Based on the information in $\mathcal{G}_i^k$, agent i maintains the joint belief

$$\pi_i^k(\theta) = \mathrm{P}\big(\Theta_i(k) = \theta \,\big|\, \mathcal{G}_i^k\big). \tag{10}$$

Two useful marginals of the joint belief are

$$\begin{aligned} \mu_{ij}^k(z) &= \mathrm{P}\big(Z_j(k) = z \mid \mathcal{G}_i^k\big), \\ \beta_i^k(h) &= \mathrm{P}\big(\eta = h \mid \mathcal{G}_i^k\big). \end{aligned} \tag{11}$$

Here, μ_{ij}^k represents agent i's belief about teammate j's mission-role state, while β_i^k represents its belief about the latent task variable. These marginals are useful for interpretation, whereas π_i^k retains statistical dependence among hidden variables when such dependence affects the decision.

The belief is converted into physical coordination through three local decision interfaces. Agent i first selects its next mission and role according to

$$(q_i^+, r_i^+) = \delta_i\big(q_i(k), r_i(k), \pi_i^k, K_i\big), \tag{12}$$

where δ_i denotes the local mission-role decision rule. A reference generator then constructs a feasible physical reference

$$\xi_i^{\mathrm{ref}} = \Gamma_i\big(x_i(k), q_i^+, r_i^+, \pi_i^k, K_i\big), \tag{13}$$

and the control input is produced by

$$u_i(k) = \kappa_i\big(x_i(k), \xi_i^{\mathrm{ref}}\big), \tag{14}$$

where Γ_i and κ_i denote the reference-generation and feedback-control maps, respectively. Hence, observed teammate behavior influences physical coordination only through the chain from inference to mission or role selection and finally to execution.

The resulting closed-loop behavior is evaluated by a task-dependent team cost J_{team}, with lower values indicating better team performance. The specific cost functions used in the two instantiations are introduced together with their respective models.

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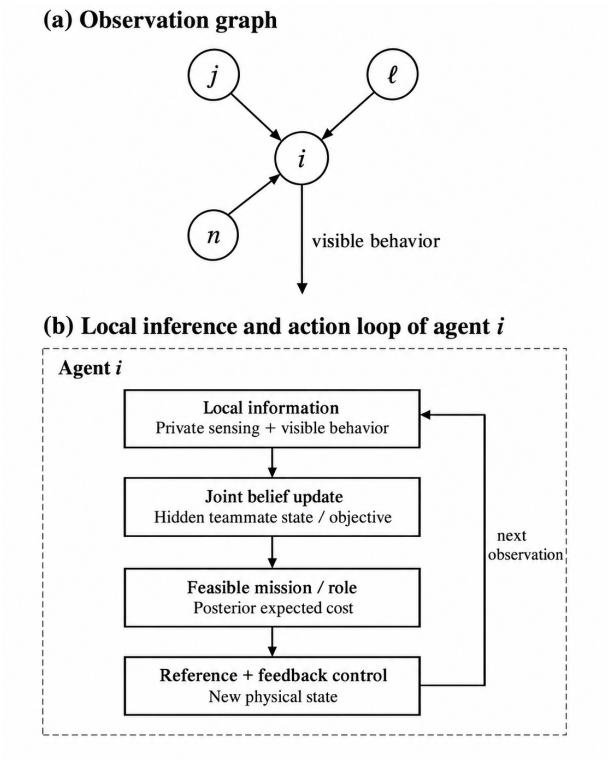


Fig. 1: Behavior-mediated multi-agent coordination framework. An agent combines private sensing with observable teammate behavior, updates a joint hidden-state belief, selects a feasible mission and role, and executes the resulting physical reference.

3. RELATIONAL INFERENCE AND ACTION VALUE

3.1. Football-Inspired Relational Model

We consider a local football-inspired coordination episode with two candidate support lanes centered around a reference point p_0. Let $e_\perp$ denote the lateral unit direction and let $d > 0$ denote the lateral lane offset, so that the two candidate support positions are $p_0 \pm de_\perp$. A hidden local geometry W induces a lane-preference difference $\Delta(W)$, and we define the latent tactical opportunity as

$$\eta = \mathrm{sign}(\Delta(W)) \in \{-1, +1\}. \tag{15}$$

Thus, $\eta = +1$ and $\eta = -1$ indicate that the positive and negative lateral lanes, respectively, are preferred for the local coordination task. The observer does not directly observe η. In this scenario, the local agent must use its own observations to infer the latent tactical opportunity.

Two visible teammates participate in the diagnostic interaction. The first teammate, termed the *initiator*, selects a fair lateral direction $S \in \{-1, +1\}$ with $\mathrm{P}(S = +1) = \mathrm{P}(S = -1) = 1/2$, independently of η. The second teammate, termed the *responder*, has a private cue $Q \in \{-1, +1\}$ satisfying $\mathrm{P}(Q = \eta \mid \eta) = 1 - \epsilon_0$ and $\mathrm{P}(Q = -\eta \mid \eta) = \epsilon_0$, where $0 \le \epsilon_0 < 1/2$.

During a diagnostic interval, the initiator produces an ideal lateral displacement mS, where $m > 0$ denotes the displacement magnitude. After observing the initiation, the responder mirrors the motion when $Q = +1$ and counter-runs when $Q = -1$, producing the ideal lateral displacement mSQ. The observer measures

$$H_1 = mS + \nu_1, \qquad H_2 = mSQ + \nu_2, \tag{16}$$

where H_1 and H_2 are the observed lateral displacements of the initiator and responder, respectively. The measurement noises ν_1 and ν_2 are modeled as independent zero-mean Gaussian random variables with variances σ_1^2 and σ_2^2.

No teammate transmits S, Q, or η to the observer. The responder's private cue affects only its physical reaction to the initiator, while the observer sees the resulting motion pair (H_1, H_2). Consequently, information about η may appear in the relation between the two observed behaviors even when neither trajectory is informative by itself.

3.2. Relational Identifiability

Let $\phi_\sigma(y - a)$ denote the probability density of $\mathcal{N}(a, \sigma^2)$ evaluated at y. Since the initiation direction S is fair and independent of η, the conditional density of H_1 is

$$p(h_1 \mid \eta) = \frac{1}{2}\phi_{\sigma_1}(h_1 - m) + \frac{1}{2}\phi_{\sigma_1}(h_1 + m), \tag{17}$$

which is identical for $\eta = +1$ and $\eta = -1$. Hence, the initiator's motion alone does not distinguish the two tactical opportunities under the symmetric model.

The same ambiguity holds for the responder when it is observed in isolation. Although Q is correlated with η, multiplication by the independent fair variable S makes SQ equally likely to be $+1$ or -1 under either value of η. Therefore,

$$p(h_2 \mid \eta) = \frac{1}{2}\phi_{\sigma_2}(h_2 - m) + \frac{1}{2}\phi_{\sigma_2}(h_2 + m), \tag{18}$$

which is again independent of η.

Marginalizing the unobserved initiation direction S and private cue Q gives the joint likelihood

$$\begin{aligned} f_\eta(h_1, h_2) = \frac{1}{2} \sum_{s \in \{-1,+1\}} & \phi_{\sigma_1}(h_1 - ms) \\ & \times \Big[(1 - \epsilon_0)\phi_{\sigma_2}(h_2 - ms\eta) \\ & \quad + \epsilon_0 \phi_{\sigma_2}(h_2 + ms\eta)\Big]. \end{aligned} \tag{19}$$

The summation accounts for the unobserved initiation side S, while the two terms in brackets correspond to a correct and an incorrect private cue Q, respectively. Unlike the two marginal densities, $f_\eta(h_1, h_2)$ retains the dependence created by the common initiation variable S. In particular, the two observations are not conditionally independent given η, because both depend on the same hidden initiation variable S.

The cross moment reveals this dependence on η. From (16),

$$\mathrm{E}[H_1 H_2 \mid \eta] = m^2 \mathrm{E}[Q \mid \eta] = m^2(1 - 2\epsilon_0)\eta, \tag{20}$$

where the zero-mean independent observation noises remove all cross terms and $S^2 = 1$. Therefore, if $m > 0$ and $\epsilon_0 < 1/2$, the cross moment is nonzero and changes sign with η. The two joint conditional distributions must consequently be different.

Theorem 1 (Relational identifiability) Consider the symmetric model above with observations in (16). Assume that S is fair and independent of (η, Q), and that ν_1 and ν_2 are independent zero-mean observation noises, independent of (S, Q, η). Then each of H_1 and H_2 individually has a conditional distribution independent of η. If $m > 0$ and $0 \le \epsilon_0 < 1/2$, their joint conditional distributions under $\eta = +1$ and $\eta = -1$ are distinct.

Proof. Because S is fair and independent of η, $H_1 = mS + \nu_1$ has the same symmetric two-Gaussian mixture distribution under both hypotheses. Likewise, for either value of η, the product SQ is fair on $\{-1, +1\}$, so H_2 has the same marginal distribution under both hypotheses, as shown in (17) and (18). However, (20) gives

$$\mathrm{E}[H_1 H_2 \mid \eta] = m^2(1 - 2\epsilon_0)\eta,$$

which is nonzero and has opposite signs for $\eta = +1$ and $\eta = -1$. Hence the two joint conditional distributions cannot be identical. □

For a transparent discrete representation of this relational signal, define the observed motion signs

$$B_j = \mathrm{sign}(H_j), \qquad j \in \{1, 2\}. \tag{21}$$

Let e_j denote the probability that measurement noise reverses the sign of the corresponding ideal displacement. Under the Gaussian observation model,

$$e_j = \Phi\left(-\frac{m}{\sigma_j}\right), \qquad j \in \{1, 2\}, \tag{22}$$

where $\Phi(\cdot)$ is the standard-normal cumulative distribution function.

The responder's cue error and its observation error combine into

$$\widetilde{e}_2 = \epsilon_0 + e_2 - 2\epsilon_0 e_2, \qquad \delta = e_1 + \widetilde{e}_2 - 2e_1\widetilde{e}_2, \tag{23}$$

where $\widetilde{e}_2$ is the effective sign error of the responder relative to $S\eta$, and δ is the error probability of the joint motion relation defined below.

Define the relational motion feature

$$C = B_1 B_2 \in \{-1, +1\}. \tag{24}$$

In the absence of cue and measurement errors, $B_1 = S$ and $B_2 = S\eta$, so their product directly recovers the latent opportunity, $C = \eta$. For $c, h \in \{-1, +1\}$, the resulting binary channel satisfies

$$\mathrm{P}(C = c \mid \eta = h) = \begin{cases} 1 - \delta, & c = h, \\ \delta, & c = -h. \end{cases} \tag{25}$$

For $0 < \delta < 1/2$, the two sign-pair distributions are statistically distinguishable. In particular,

$$D_{\mathrm{KL}} = (1 - 2\delta)\log\left(\frac{1-\delta}{\delta}\right) > 0, \tag{26}$$

where D_{KL} denotes the Kullback–Leibler divergence between the distributions of (B_1, B_2) under $\eta = +1$ and $\eta = -1$. Thus, even the one-bit relational feature C retains information about the latent opportunity.

For the base parameters used later in the simulation, $e_1 = 0.08$, $e_2 = 0.12$, and $\epsilon_0 = 0.10$, giving $\widetilde{e}_2 = 0.196$ and

$$\delta = 0.24464, \qquad \mathrm{P}(C = \eta) = 1 - \delta = 0.75536. \tag{27}$$

This provides the analytical reference for the finite-sample relational accuracy reported in Sec. 4.

The individual-marginal ambiguity above relies on the symmetric model: if the observer independently learns S, or if visibility selectively reveals the hidden variables, a single observed trajectory may itself become informative.

3.3. Mission and Action Value

We next connect relational inference to the observer's own decision. When agent i undertakes COVER, it selects a lateral support role $a \in \{-1, +1\}$, where the two values correspond to the positive and negative $e_\perp$ directions, and tracks the reference $p_i^{\mathrm{ref}} = p_0 + ade_\perp$, where p_i^{ref} is the positional reference for agent i, p_0 is the local reference point defined in Sec. III-A, and $d > 0$ is the lateral lane offset. To quantify whether the selected role covers the appropriate lane, define the lateral task loss

$$J_{\mathrm{lane}}(a, \eta) = d^2(a - \eta)^2. \tag{28}$$

Since $a, \eta \in \{-1, +1\}$, the loss is zero when the selected support side agrees with the latent opportunity and equals $4d^2$ when the wrong side is selected. This loss measures spatial coverage error rather than disagreement between abstract role labels.

Assume a fair prior, $\mathrm{P}(\eta = +1) = \mathrm{P}(\eta = -1) = 1/2$. Without teammate-behavior information, either fixed support role has error probability $1/2$, and hence the minimum expected lane loss under private information is $2d^2$. After observing the relational feature C, choosing $a = C$ has error probability δ by (25). The corresponding expected lane loss is therefore $4d^2\delta$. Since $\delta < 1/2$, relational behavior strictly improves the expected role-level loss whenever the COVER mission is undertaken.

Role improvement does not by itself imply that the observer should undertake the coverage mission. We therefore make the mission decision explicit by allowing

$$q_i^+ \in \{\mathrm{WAIT}, \mathrm{COVER}\}.$$

Let $h > 0$ denote the task cost of remaining in WAIT, representing the loss associated with delaying or omitting the current support opportunity. Entering COVER incurs a fixed movement or switching charge $c \ge 0$ in addition to the spatial loss $J_{\mathrm{lane}}(a, \eta)$ in (28). The resulting optimal expected mission losses are

$$R_{\mathrm{priv}} = \min\{h,\ c + 2d^2\}, R_{\mathrm{rel}} = \min\{h,\ c + 4d^2\delta\}, \tag{29}$$

where R_{priv} denotes the minimum expected loss using private information only, and R_{rel} denotes the minimum expected loss after conditioning on the relational motion feature.

Theorem 2 (Observation-induced mission transition) Assume the fair prior $\mathrm{P}(\eta = +1) = \mathrm{P}(\eta = -1) = 1/2$, both support roles are feasible, and $0 < \delta < 1/2$. The optimal private-information and relational-information mission risks are given by (29). Moreover,

$$R_{\mathrm{rel}} < R_{\mathrm{priv}} \quad \Longleftrightarrow \quad h > c + 4d^2\delta. \tag{30}$$

In particular, if

$$c + 4d^2\delta < h < c + 2d^2, \tag{31}$$

the private-information decision is WAIT, whereas the behavior-informed decision is COVER with role $a = C$.

Proof. Under private information and the fair prior, either support side is incorrect with probability $1/2$. Thus, the expected COVER loss is $c + 2d^2$, while WAIT costs h. After observing C, the role $a = C$ is incorrect with probability δ, so the expected COVER loss becomes $c+4d^2\delta$. This proves (29). Since $\delta < 1/2$, $c + 4d^2\delta < c + 2d^2$. Hence the two optimal risks are equal when $h \le c+4d^2\delta$ and satisfy $R_{\mathrm{rel}} < R_{\mathrm{priv}}$ otherwise. Under (31), WAIT is strictly optimal without relational information, whereas COVER is strictly optimal after observing it. □

Theorem 2 separates information from decision value: a more informative observation is useful here because it crosses a task-dependent decision boundary and changes the observer's optimal mission, not merely because it improves classification accuracy.

For the base setting used in Sec. IV, $d = 1$, $c = 0.1$, $h = 1.5$, and $\delta = 0.24464$. The private COVER risk is therefore 2.1, so the private-information policy selects WAIT. In contrast, the relational COVER risk is 1.07856, which induces the COVER mission and yields an analytical task-risk reduction of 0.42144. These values are model expectations; the finite-sample and closed-loop results are reported separately in Sec. IV.

The decision-level gain must still survive physical execution. Suppose the behavior-informed COVER action reaches terminal lateral displacement $z_i = Cd + e_i$ with $|e_i| \le \bar{e}$, where z_i is measured relative to p_0, e_i denotes the terminal tracking error, and $\bar{e} \ge 0$ is a valid tracking-error bound. Let $\zeta \ge 0$ bound the additional team-cost difference caused by effects not included in the lane loss, such as teammate response and model mismatch. Under the same comparison horizon,

$$\begin{aligned} J_{\mathrm{team}}^{\mathrm{WAIT}} - J_{\mathrm{team}}^{\mathrm{COVER}} \ge h - c - 4d^2\delta \\ - 4d\bar{e} - \bar{e}^2 - \zeta. \end{aligned} \tag{32}$$

Thus, the relational decision retains a team-level advantage when its modeled gain exceeds the execution and model-error terms. The closed-loop experiments in Sec. IV evaluate the realized cost directly.

4. CLOSED-LOOP EVALUATION AND EXPERIMENTS

4.1. Football-Inspired Coordination

The football-inspired study evaluates the inference-to-action chain at two successive levels. First, we sample the symmetric generative model of Sec. 3 to test whether individual teammate motions remain ambiguous while their relation reveals the latent opportunity η, and whether this additional information changes the observer's mission decision. Second, we embed the same decision logic in a planar three-agent simulation with bounded-speed and bounded-acceleration motion, and evaluate whether the resulting decision advantage persists after physical execution.

We compare several information structures using the same task-loss definition. PRIVATE uses no teammate-behavior evidence and acts from the private prior. H1 and H2 use only the corresponding single observed motion. SIGN uses the relational feature $C = \mathrm{sign}(H_1)\mathrm{sign}(H_2)$, whereas CONTINUOUS evaluates the joint likelihood in (19) using the full pair (H_1, H_2). ORACLE is an additional-information reference that observes the true value of η. All non-oracle policies select WAIT or a lateral COVER role by comparing the corresponding posterior expected task losses.

Figure 2 summarizes the generator-level evidence. The simulation uses the normalized case $m = 1$; the noise terms denoted by ϵ_1 and ϵ_2 in the schematic correspond to the observation noises ν_1 and ν_2 in Sec. III. Importantly, S is sampled independently of (η, Q), and the private cue Q affects the responder through the physical relation SQ rather than the initiator's choice of S. Panels (b) and (c) show the resulting contrast between an uninformative individual marginal and the objective-dependent joint motion distribution.

In the base generator experiment with 100,000 samples, the H1 and H2 policies each achieved classification accuracy 0.50007, consistent with the η-independent marginals established in Theorem 1. In contrast, the SIGN relation achieved accuracy 0.75548. This closely matches the analytical prediction $1 - \delta = 0.75536$ obtained from (27). The empirical relation-error probability was therefore 0.24452, compared with the theoretical value $\delta = 0.24464$.

The same experiment confirms that relational information has decision value under the task loss in Sec. III-C. Mean losses were 1.500000 (PRIVATE), 1.078080 (SIGN), 1.005784 (CONTINUOUS), and 0.100000 (ORACLE). CONTINUOUS chose COVER in $\sim 76.2\%$ of samples, defaulting to WAIT when posterior evidence was insufficient—demonstrating that the loss reduction stems not from classification alone, but from belief-driven mission decisions and associated expected task loss, as formalized in Theorem 2.

We next evaluate the same inference-to-action mechanism under physical motion. Each episode involves an initiator, a responder, and an observer. During the diagnostic phase ($t \le 5$ s), the initiator and responder execute

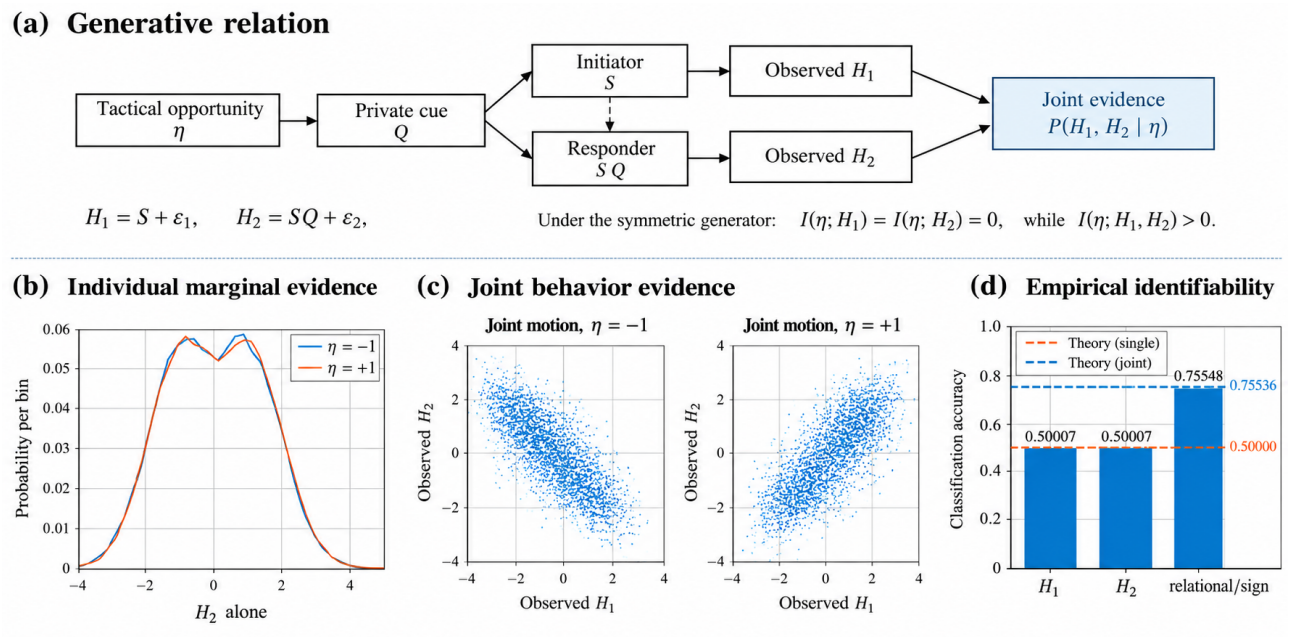


Fig. 2: Relational identifiability in the football-inspired generator. (a) Schematic generative relation in the normalized case $m = 1$; S is independent of (η, Q), while Q affects the responder through SQ. (b) A representative individual marginal is invariant with respect to η. (c) The joint motion pair exhibits opposite relational structure under the two hypotheses. (d) Empirical classification accuracy compared with the analytical single-motion and relational predictions.

prescribed lateral maneuvers under bounded dynamics. At $t = 5$ s, the observer infers their lateral displacements, updates its posterior once, and selects WAIT, LEFT, or RIGHT. The observer then tracks the chosen reference for the remaining 3 s. Thus, the experiment tests a single observation–decision–execution cycle—not continuous online Bayesian filtering.

Figure 3 illustrates one complete observation–decision–execution episode; aggregate performance is reported below.

We evaluate 200 matched scenarios under each of three sensing conditions: base, high noise, and 40% trajectory missingness. All policies share the same scenario realization; comparisons are therefore paired. H1 and H2 are omitted from Table I because their decisions are essentially identical to PRIVATE in the symmetric model.

The executed team cost extends the decision loss used in Sec. III-C with physical-motion terms. In the normalized setting $d = 1$, WAIT has terminal task cost 1.5, whereas a COVER action has terminal cost $0.1+(z_i-\eta)^2$, where z_i denotes the observer's terminal lateral position relative to p_0. The implemented team cost further includes 0.01 times the total distance traveled by the three agents and a prespecified penalty for pairwise collisions. Hence, the physical evaluation measures the realized consequence of both the mission decision and its execution rather than the posterior loss alone. For CONTINUOUS, the likelihood of (19) is applied to the measured finite-motion displacements. It therefore serves as the ideal-model posterior used by the controller rather than an exact likelihood of the full physical simulator.

Table 1 shows that the decision-level gain survives

Table 1: Mean executed team cost in the football experiment (lower is better).

Condition	PRIVATE	SIGN	CONT.	ORACLE
Base	1.5208	1.1648	1.0328	0.1312
High noise	1.5208	1.5208	1.5058	0.1312
40% missing	1.5208	1.4664	1.4160	0.1312

physical execution in the base setting. CONTINUOUS reduces the mean team cost from 1.5208 under PRIVATE to 1.0328. Using the matched scenarios, the paired improvement $J_{\text{PRIVATE}} - J_{\text{CONTINUOUS}}$ is 0.48803, with a 95% bootstrap confidence interval $[0.29289, 0.67622]$. SIGN also yields a paired improvement of 0.35597, with interval $[0.09250, 0.59917]$. These results provide closed-loop evidence that the relational information and mission transition from Sec. III reduce realized team cost under bounded physical execution.

The gain decreases as relational evidence degrades. Under high observation noise, CONTINUOUS has mean cost 1.5058, only 0.0150 below PRIVATE, with a paired 95% interval $[-0.1066, 0.1308]$. With 40% trajectory missingness, its mean cost is 1.4160, and the mean paired improvement is approximately 0.1048; the corresponding confidence interval also includes zero. This behavior is consistent with the theoretical separation between information availability and action value: as the relational evidence becomes less reliable or unavailable, fewer episodes provide sufficient evidence to justify a beneficial mission change.

Taken together, the generator-level and physical experiments provide empirical checks of distinct parts of the proposed chain. The first is consistent with the relational identifiability of Theorem 1 and the mission-level deci-

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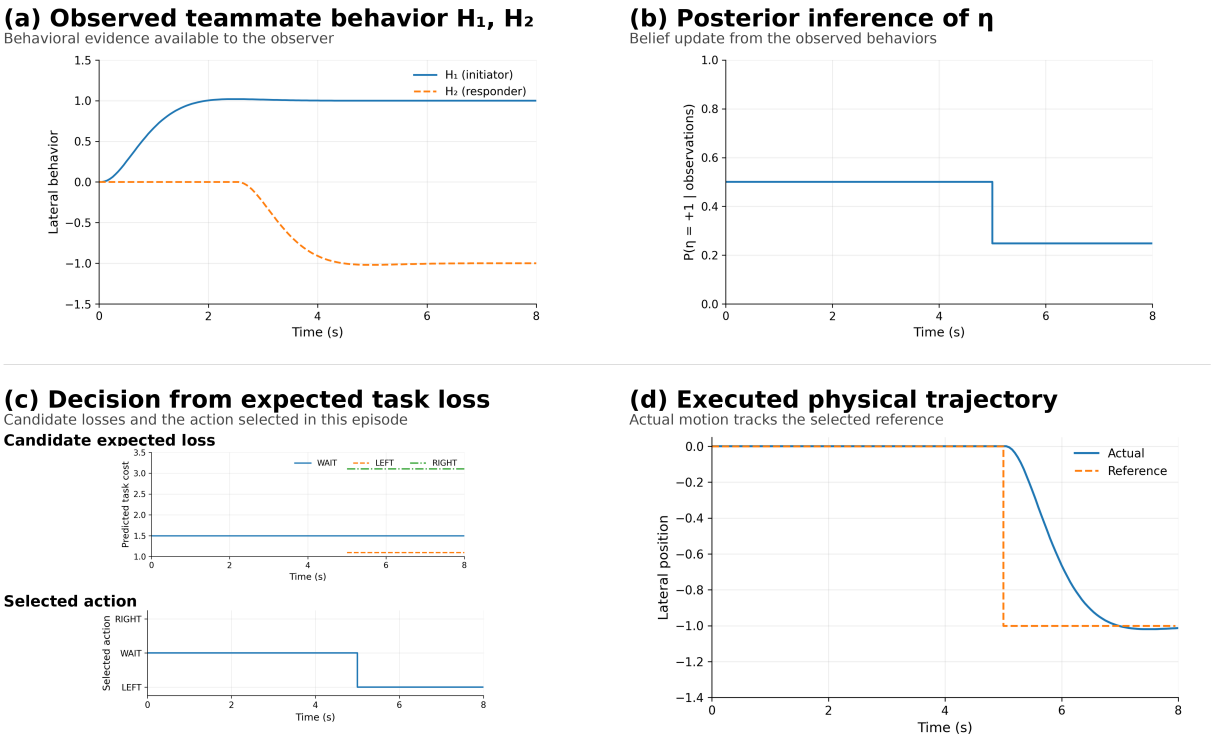


Fig. 3: Representative football inference-to-execution episode. (a) Observed teammate lateral motions during the diagnostic phase; the terminal displacements at $t = 5$ s define H_1 and H_2. (b) The corresponding posterior over η. (c) Posterior expected task losses and the selected action. (d) Reference and executed lateral trajectory after the decision. The posterior is updated once at the end of the diagnostic phase.

sion value of Theorem 2. The second shows that this decision advantage can persist after reference tracking and physical motion, while the realized gain depends on the quality of the available behavioral evidence.

4.2. Multi-Agent Formation Recovery

We next examine whether the same behavior-to-action principle extends to a larger decentralized closed loop. Ten agents must recover a formation consisting of target slots $g_s \in \mathbb{R}^2$, $s \in \{1, \dots, N\}$, with $N = 10$. For each visible teammate j, agent i maintains a local distribution $p_i^j(s)$ over the teammate's current target slot. The distribution is updated from observed position and velocity direction, with uncertainty increased when the teammate is not visible. This estimator is a tractable motion-based approximation of the general inference interface in Sec. II, rather than an exact joint Bayesian filter.

For each candidate slot s, the visible-neighbor beliefs are combined into the occupancy score $\mathrm{occ}_i(s) = 1 - \prod_{j \in \mathcal{N}_i^o} [1 - p_i^j(s)\rho_{ij}(s)]$, where $\mathcal{N}_i^o$ is the current observable-neighbor set and $\rho_{ij}(s) \in [0, 1]$ is a priority factor determined by the relative distances of agents i and j to slot s. Agent i ranks candidate slots using $\mathrm{score}_i(s) = \|x_i - g_s\|/v_{\max} + 8\,\mathrm{occ}_i(s) + 0.3\,\mathbf{1}\{s \neq r_i\}$, where $v_{\max}$ is the speed limit and r_i is the currently selected slot. The occupancy term is a local aggregation score rather than a calibrated joint probability. Hysteresis and a minimum dwell time suppress frequent switching, and the selected slot is tracked using a bounded local velocity controller with short-range collision avoidance.

The primary comparisons are against REACTIVE, which records a visible teammate's slot only after the teammate is nearly in place, and POSITION, which uses neighbor position but not motion direction. The latter is a prespecified strong baseline. Team formation quality is measured by $D(t) = \min_{a \in \mathcal{A}} \sum_{i=1}^{N} \|x_i(t) - g_{a(i)}\|^2$, where $\mathcal{A}$ is the set of one-to-one assignments from agents to target slots. The reported team cost is

$$J = 5D_{\mathrm{f}} + 0.15 \int_{0.15}^{24} D(t)\, dt + 20N_{\mathrm{col}} + 0.02L_{\mathrm{path}} + 0.05N_{\mathrm{repl}} + 5(N - N_{\mathrm{fill}}),$$

where D_{f} is the terminal assignment error, N_{col} the number of simulation steps containing a near-distance collision event, L_{path} the total path length, N_{repl} the number of role replans, and N_{fill} the final number of occupied slots.

We evaluate 400 matched scenarios: 200 ring initializations and 50 each for two-row, asymmetric, occluded, and private-goal settings. All policies share the same initial condition, noise, and occlusion stream within each scenario. Paired differences are defined as $\Delta J = J_{\mathrm{baseline}} - J_{\mathrm{MULTI}}$, so positive values favor MULTI; 95% intervals use 2000 scenario-level bootstrap resamples.

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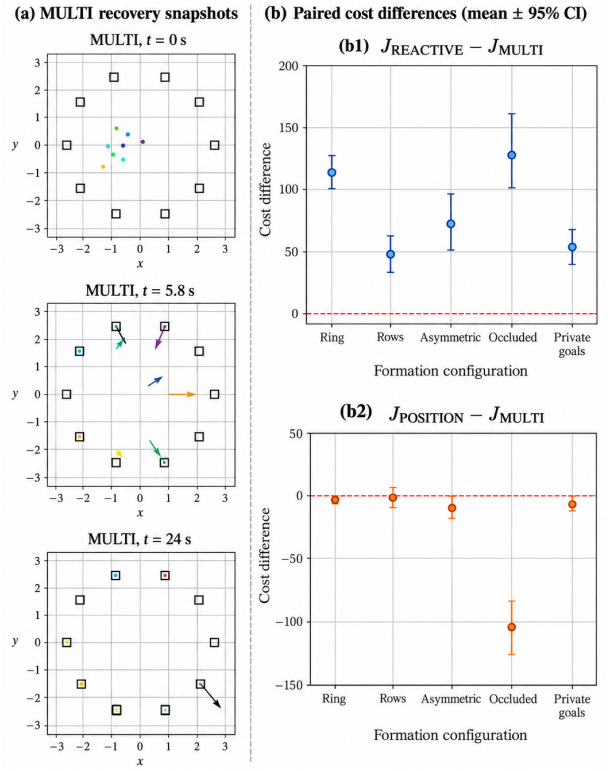


Fig. 4: Multi-agent formation recovery and paired cost comparisons. (a) MULTI recovery in a fixed ring scenario. (b1) Paired differences $J_{\mathrm{REACTIVE}} - J_{\mathrm{MULTI}}$ and (b2) $J_{\mathrm{POSITION}} - J_{\mathrm{MULTI}}$ across five formation configurations. Markers denote mean paired differences and error bars show 95% bootstrap intervals. Positive values indicate a lower team cost for MULTI.

Figure 4(b1) shows a consistent advantage over REACTIVE: all five paired intervals are above zero. The mean improvement is 112.702 in the ring setting (95% CI $[101.914, 124.324]$) and 128.659 under occlusion ($[98.321, 157.995]$), indicating a clear benefit over delayed response.

The comparison with POSITION is different. In the ring setting, $J_{\mathrm{POSITION}} - J_{\mathrm{MULTI}} = -1.465$ (95% CI $[-1.841, -1.070]$), and the asymmetric, occluded, and private-goal settings show the same direction. The two-row interval $[-1.564, 0.100]$ crosses zero. Thus, motion-based inference improves over delayed reaction but does not provide additional closed-loop value beyond the strong position-only cue in these tests.

5. DISCUSSION AND CONCLUSION

This work studies how neighboring agents' actions reveal hidden states and decision-relevant information unavailable from local measurements. In the football-inspired model, individually ambiguous motions become informative through their relation, changing the observer's mission and role while retaining value after physical execution. Formation recovery shows the complementary limit: anticipatory reasoning improves over delayed reaction, but richer inference adds no closed-loop value when a simpler cue already supports the downstream decision.

These results motivate viewing physical behavior as an endogenous information channel in multi-agent systems, complementary to broader local-interaction and nature-inspired coordination mechanisms [10]. Such information can reshape an agent's own decision rule and connect behavior inference with distributed estimation, role allocation, and networked control. Future work will study richer hidden states, asynchronous reasoning, and conditions for safe, scalable, and effective collective behavior in multi-agent and swarm systems.

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